PEER-REVIEWED RESEARCH DOI: 10.1038/s41586-026-08112-x

High-Throughput Whole-Genome Sequencing and De Novo Mutation Landscape in Chironomus riparius

Burak Bulut, Ph.D. [1,2,*], Elena Rostova, M.Sc. [1], Marcus Vance, Ph.D. [3]

1. Department of Evolutionary Genomics, Max Planck Institute for Biology, Tubingen, Germany
2. Computational Biology & Systems AI Laboratory, Goethe University, Frankfurt am Main, Germany
3. European Molecular Biology Laboratory (EMBL), Heidelberg, Germany. *Email: burak.buluut@outlook.com

ABSTRACT

We investigate the de novo germline mutation rate and somatic variant accumulation in Chironomus riparius under acute heavy metal environmental selection pressure. Using whole-genome sequencing (WGS) on the Illumina NovaSeq 6000 platform with 60x coverage across multi-generational lineages, we identified 1,428 high-confidence single nucleotide variants (SNVs) and 184 small insertions/deletions. High-throughput bioinformatic processing was orchestrated through GATK4 HaplotypeCaller, BWA-MEM read alignment, and rigorous quality filtering (MAPQ >= 40, DP >= 25, QD < 2.0). Our results show an elevated transition-to-transversion (Ti/Tv) ratio of 2.34 and an estimated spontaneous mutation rate of 3.12 x 10^-9 per site per generation. Differential gene expression (RNA-seq) revealed coordinated upregulation of heat shock protein families (Hsp70, Cyp450). This study provides foundational genomic resources and benchmark mutation rates for environmental sentinel dipterans.

Keywords: Chironomus riparius, whole-genome sequencing (WGS), de novo mutation, heterozygosity, variant calling, GATK, Illumina, FASTQ.

1. INTRODUCTION

Understanding the baseline rates of spontaneous mutations and how environmental stressors perturb genomic stability is a central question in evolutionary genetics and ecotoxicology. Non-biting midges of the genus Chironomus, particularly Chironomus riparius, are ubiquitous freshwater macroinvertebrates widely utilized as bioindicators of sediment toxicity and aquatic environmental quality. Despite extensive phenotypic characterization, the precise rate of de novo germline mutations and their distribution across the nuclear genome have remained poorly quantified due to the historical scarcity of high-continuity reference assemblies.

Next-generation high-throughput sequencing has fundamentally transformed non-model genomics. By leveraging deeply sequenced parent-offspring trios and multi-generational lineages, it is now feasible to detect true de novo mutations with high statistical specificity while mitigating false discovery rates caused by sequencing errors or PCR amplification artifacts. In this investigation, we established a rigorous variant calling pipeline conforming to GATK best practices, coupled with automated quality control and high-performance computing (HPC) workflows.

2. MATERIALS AND METHODS

2.1 Biological Sampling and Colony Maintenance

Chironomus riparius egg clutches were sampled from uncontaminated reference sites in Central Europe and reared under controlled climate chamber conditions (20 +/- 1 deg C, 16:8 light:dark photoperiod). Five independent pedigree lineages were maintained across 10 generations under standard laboratory conditions (Lineages L01 to L05), while two parallel lineages were subjected to sublethal cadmium acetate exposure (0.5 ug/L) to simulate persistent environmental stress.

2.2 High-Throughput DNA Extraction and Illumina Sequencing

High-molecular-weight genomic DNA was extracted from individual adult specimens using the Qiagen MagAttract HMW DNA Kit. Paired-end sequencing libraries with an average insert size of 380 bp were prepared using the Illumina TruSeq DNA PCR-Free protocol to minimize library amplification bias. Deep sequencing was conducted on an Illumina NovaSeq 6000 sequencer, yielding an average of 42.5 million 150 bp paired-end read pairs per individual (approximately 60x nominal coverage).

2.3 Computational Pipeline, Alignment, and Variant Calling

Raw FASTQ files were subjected to automated quality control using fastp (v0.23.4) with parameters --detect_adapter_for_pe and --qualified_quality_phred 30. High-quality filtered reads were aligned to the chromosome-level Chironomus riparius reference genome assembly (v2.1) using BWA-MEM (v0.7.17-r1188). Alignment files in BAM format were sorted and duplicates marked via Picard Tools (v3.1.1). Mapping quality scores (MAPQ) were evaluated across all genomic regions.

Variant calling was executed using GATK4 HaplotypeCaller (v4.5.0) in GVCF mode across each lineage cohort, followed by joint genotyping via GenotypeGVCFs. Stringent variant quality score recalibration and hard filtering thresholds were applied: Quality by Depth (QD >= 2.0), Fisher Strand (FS <= 60.0), Mapping Quality (MQ >= 40.0), and ReadPosRankSum >= -8.0. Candidate de novo mutations were confirmed by checking that alternate alleles were absent in both parental samples and present in at least 25% of reads in the focal progeny with a minimum read depth (DP) of 25x.

2.4 High-Performance Cluster Orchestration and SLURM Architecture

All analytical steps were modularized into reproducible Snakemake workflows deployed on an institutional SLURM HPC cluster. Resource allocations were monitored in real time using PipeWatch telemetry to guarantee deterministic resource bounds, while sample provenance and checksum verifications were enforced via SampleSheriff CLI checkpoints.

3. RESULTS

3.1 Genome-Wide Sequencing Depth and Variant Calling Summary

Across all sequenced individuals, we achieved an average mapping rate of 98.4% and an effective mean coverage depth of 58.2x. In total, 1,428 high-confidence de novo single nucleotide variants were validated across 10 generations. Table 1 summarizes the sequencing yield, alignment efficiency, transition-to-transversion ratios, and estimated heterozygosity.

Table 1. Whole-genome sequencing yield, alignment metrics, and variant statistics across Chironomus lineages.

Lineage ID	Reads (M)	Mapped (%)	Depth (X)	Ti/Tv Ratio	De Novo SNVs	Heterozygosity
L01 (Control)	44.2	98.7%	62.4x	2.31	284	0.0078
L02 (Control)	41.8	98.5%	59.1x	2.35	276	0.0075
L03 (Control)	43.1	98.3%	60.8x	2.28	291	0.0081
L04 (Cadmium Stress)	46.0	97.9%	64.9x	2.44	398	0.0094
L05 (Cadmium Stress)	42.9	98.1%	60.5x	2.41	379	0.0092

3.2 Mutation Rate Spectrum and Environmental Impact

The overall spontaneous de novo mutation rate in unexposed control lineages was estimated at mu = (3.12 +/- 0.28) x 10^-9 per base pair per generation. In contrast, lineages subjected to chronic cadmium exposure exhibited a statistically significant 38% elevation in germline mutation frequency (mu = 4.31 x 10^-9, Student t-test p = 0.00034). Analysis of the mutational spectrum revealed a preponderance of C:G -> T:A transitions, characteristic of oxidative DNA damage and cytosine deamination.

3.3 Differential Gene Expression in Stress Response Pathways

RNA-seq profiling of whole larvae under identical treatment regimens demonstrated significant differential gene expression (FDR < 0.01) across 412 genes. Transcripts encoding heat shock protein 70 (Hsp70) and cytochrome P450 monooxygenases (Cyp450) showed >4.5-fold induction in cadmium-exposed lines, correlating with localized somatic mutation hotspots.

4. DISCUSSION & REPRODUCIBILITY

Our empirical quantification of the Chironomus riparius mutation rate places this sentinel species within the evolutionary envelope observed for Diptera, intermediate between Drosophila melanogaster (2.8 x 10^-9) and Heliconius melpomene (2.9 x 10^-9). The marked acceleration of point mutations under environmental xenobiotic exposure underscores the sensitivity of eukaryotic mismatch repair mechanisms to toxic heavy metal ions.

From a computational methodology perspective, non-model genomic investigations frequently grapple with fragmented assemblies and elevated heterozygosity that generate false-positive structural variant calls. By strictly enforcing joint genotyping, high mapping quality thresholds (MAPQ >= 40), and independent biological replicate cohorts, we eliminated over 94% of spurious candidate loci. The integration of containerized workflows and real-time execution monitoring ensures complete auditability and compliance with FAIR (Findable, Accessible, Interoperable, and Reusable) scientific principles.

5. REFERENCES

[1] Bulut, B., Oppold, A., & Pfenninger, M. (2024). Ecotoxicogenomics of non-biting midges. Environmental Pollution, 312, 119842.
[2] Li, H., & Durbin, R. (2009). Fast and accurate short read alignment with Burrows-Wheeler transform. Bioinformatics, 25(14), 1754-1760.
[3] McKenna, A., et al. (2010). The Genome Analysis Toolkit: a MapReduce framework for analyzing next-generation DNA sequencing data. Genome Research, 20(9), 1297-1303.
[4] Pfenninger, M., et al. (2022). Chromosome-scale reference genome of Chironomus riparius. BMC Genomics, 23, 412.
[5] Keightley, P. D., et al. (2014). Analysis of the genome sequences of three Drosophila melanogaster laboratory strains. Genetics, 196(1), 313-323.
[6] Chen, S., et al. (2018). fastp: an ultra-fast all-in-one FASTQ preprocessor. Bioinformatics, 34(17), i884-i890.
[7] Oppold, A. M., et al. (2015). Epigenetic regulation of temperature adaptation in Chironomus riparius. Molecular Ecology, 24(12), 3020-3032.
[8] Robinson, J. T., et al. (2011). Integrative genomics viewer. Nature Biotechnology, 29(1), 24-26.

DATA & CODE AVAILABILITY

Raw sequencing reads: NCBI SRA under BioProject PRJNA984120. Variant callsets (VCFs) and Snakemake pipelines are accessible on Zenodo (10.5281/zenodo.823910) and GitHub: https://github.com/BurakBulut-sh/BioRead